

Local anchoring of global knowledge: How domestic market collaboration shapes the relationship between foreign collaboration breadth and new-to-market innovation

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ABSTRACT

This study examines whether the breadth of a firm's foreign collaboration portfolio is associated with subsequent new-to-market innovation, whether domestic market collaboration moderates that association, and whether such complementarity is stronger for commercialization than for market novelty introduction. Using panel data from the German Mannheim Innovation Panel in the period of 2019 to 2024, pooled binary logit models are estimated on 4 693 firm-wave observations from 3 519 firms. Foreign collaboration breadth establishes an inverted-U association with new-to-market innovation, peaking at around three foreign partner categories. Domestic market breadth is positively associated with new-to-market innovation on its own but interacts negatively with foreign breadth, indicating attenuation instead of complementarity while no focal term reaches significance in the conditional high-commercialization sample. Results therefore do not support the assumed geographically distant and locally anchored collaboration portfolios reinforcement towards one another, and they highlight the distinction between knowledge sourcing and commercial value capture.

1. INTRODUCTION

In an increasingly globalized world, a firm's ability to extend beyond its organizational structure became pivotal for innovation. Companies can lack specific technical knowledge, capabilities or the local know-how to identify or exploit market opportunities. This is where external partnerships come to fruition by connecting companies to specialized knowledge, thus, innovation literature considers collaboration a central mechanism of open innovation (Laursen & Salter, 2006; West & Bogers, 2014). This is especially true regarding new-to-market innovation, covering products that are to some extent novel to the specific market, and consequently firms require more than the sole use of local solutions (Hsieh et al., 2018; Kesidou et al., 2023).

The central concern is the breadth of firms' collaboration portfolios, more precisely, the diversity of collaboration partner categories that they work with. The broader a collaboration portfolio the more likely that collaborators create different synergies, reveal hidden needs, and are exposed to non-redundant, heterogenous knowledge, which allows for the recombination of these elements under innovation uncertainty (Srinivasan et al., 2021). Leiponen and Helfat (2010) suggest the concept of portfolio logic, as innovation projects come with great uncertainty, increasing the breadth of external knowledge domains can raise the probability of at least one contributing to the firm's novel innovation output. Yet the consistent broadening of collaboration partners is not without costs. According to the findings of Laursen and Salter (2006), increasing external search breadth becomes counterproductive after a certain level regarding innovation performance. Expanding breadth of partner categories bring about additional managerial, coordination, and governance demands, through which the benefits of an external knowledge source can be outweighed by the rise of costs (Audretsch & Belitski, 2023; Izotova & Bolívar Ramos, 2024).

An important condition of this tension is geographic location. Foreign collaboration breadth opens up the firm to technological advancement, market experiences, not, or less available domestically which increases the diversity of knowledge that can be exploited for innovation purposes (Hsieh et al., 2018; Kesidou et al., 2023). According to Jacob et al. (2023) concurrently more diverse collaboration portfolios are more complex to coordinate as they can create knowledge redundancy. Based on the above logic, an inverted-U shaped relationship is expected, as foreign collaboration breadth expands it may support innovation processes until it becomes counterproductive once the portfolio exceeds the firm's integration capacity.

Building on this, the core argument of this master thesis is that the value of foreign knowledge may depend on the extent of domestic market anchoring. According to Bathelt et al. (2004) the knowledge from locally established interactions, "local buzz", and the more deliberately built-up channels of distant knowledge, "global pipelines", do not function as alternatives but complements. Therefore, firms might gain a distinct benefit when local partners help translate the foreign concepts, and non-redundant ideas from global pipelines. Related literature further supports that the combination of local-global knowledge, and proximity can ease interpretation and implementation of externally gained knowledge (Un & Rodríguez, 2018; Leckel et al., 2022).

Previous research does not provide a detailed answer regarding this complementarity. Studies examined different foreign partner configurations to generate substitutions, complementarities, or even additional coordination burdens (Hsieh et al., 2018; Haus-Reve et al., 2019). Kesidou et al. (2023) further emphasizes that geography of collaboration influences the introduction of a market novelty and its commercialization differently. This distinction is especially important as commercialization from market novelty is a separate outcome, which requires additional factors like actual production, distribution and sales (West & Bogers, 2014; Salehi et al., 2026; Livieratos et al., 2026).

To fill this gap, this thesis aims to explore if foreign collaboration breadth has a nonlinear relationship with subsequent innovation output, if domestic market collaboration serves as a moderator in that relationship, and whether local anchoring has a greater influence on commercialization than introduction alone. The main goal is to examine the interaction of foreign knowledge sourcing with local market embeddedness with the following main research question: How is foreign collaboration breadth associated with subsequent new-to-market innovation, and how does domestic market collaboration condition this relationship during innovation introduction and commercialization?

There are four theoretical perspectives that the study will build upon. Firstly, organizational search theory, which defines foreign collaboration breadth as external search and motivates the expected threshold beyond which additional partner categories stop contributing to innovation (March, 1991; Laursen & Salter, 2006). Second, absorptive-capacity theory specifies the internal integration capacity needed to integrate foreign knowledge and thus explains where that threshold originates (Cohen & Levinthal, 1990; Zahra & George, 2002). Thirdly, the before mentioned local-buzz/global-pipelines perspective of Bathelt et al. (2004), which provides the reasoning for treating domestic market collaboration as a local anchor that helps translate foreign knowledge. Finally, complementary-assets theory explored by Teece (1986), which separates the introduction of a new-to-market innovation output from its commercialization output and guides the expectation that local anchoring matters more for capturing commercial value. Overall, these perspectives justify the three hypotheses developed in the following.

The thesis aims to establish three main contributions. First, extending research on external-search breadth by empirically testing a nonlinear relationship between foreign breadth and market novelty introduction supported by recent studies of limits of openness (Audretsch & Belitski, 2023; Jacob et al., 2023; Izotova & Bolívar Ramos, 2024). Second, another contribution adds to the literature on the geography of innovation by using domestic market

breadth not as a proxy for international openness but as an anchor conditioning foreign breadth (Un & Rodríguez, 2018; Leckel et al., 2022). Lastly, the study extends collaboration literature beyond the sole introduction of a market novel product with exploring its commercial significance (Kesidou et al., 2023; Salehi et al., 2026).

In the next section the theoretical framework will be developed along with the hypotheses. After, the dataset and methodology will be described in detail, followed by the result section and lastly the conclusion and discussion.

2. THEORETICAL FRAMEWORK AND HYPOTHESIS DEVELOPMENT

2.1 Foundations: Global Knowledge Sourcing and Local Anchoring

To provide a theoretical foundation for the main question of this thesis, first, external collaboration needs to be defined sufficiently. External collaboration according to previous literature is a form of organisational search and interorganisational learning. Companies search to better complete their knowledge as solely relying on familiar processes and routines can make them stuck in local solutions. Broadening the firm's collaboration partners and activities extends the knowledge domain of which the firm is exposed to, which can result in positive synergies and create opportunities to combine previously disconnected knowledge elements (March, 1991; Leiponen & Helfat, 2010). The extension of the knowledge pool is especially true for foreign collaboration types as international partners are essentially located in environments with different cultural, technological and market contexts. Such diversity has the ability to challenge established assumptions, supporting knowledge recombination foreign to the firm, but poses challenges in transferring and coordinating this knowledge (Hsieh et al., 2018; Kesidou et al., 2023; Jacob et al., 2023).

Nevertheless, accessing external knowledge is only the beginning of the process. To apply interorganisational learnings, the firm needs to identify, assimilate and apply this knowledge

for actual commercial success. While Cohen and Levinthal (1990) emphasises related prior knowledge, Zahra and George (2002) identifies two types of absorptive capacity, potential (acquisition and assimilation) and realised (transformation and exploitation). This distinction is key, as actual innovation performance depend on the company's ability to best transform the acquired knowledge into actionable strategies and carry those out successfully. As openness expands, knowledge transformation can become increasingly complex resulting in collaboration portfolios exceeding the firm's integration and implementation capacity which is supported by recent findings that benefits of open innovation is bounded by geographic and organizational complexity (Audretsch & Belitski, 2023).

The assumed complementarity between foreign collaboration and innovation outcomes anchored by domestic collaboration builds upon the local-global perspective of context dependent knowledge (Bathelt et al., 2004). Non-redundant ideas from global pipelines are meaningless unless they are adequately interpreted to fit the local market. Domestic partnerships can provide the necessary shared language, market context and institutional understanding to extend the firm's already existent capabilities to integrate foreign knowledge. Related open innovation literature argues that proximity to collaborators can facilitate trust, recurring interaction, and the application of foreign knowledge, while further evidence on the local-global collaboration complementarity suggests a possible positive association (Un & Rodríguez, 2018; Leckel et al., 2022). Thus, domestic market breadth can be best described as an anchoring mechanism for the integration of externally sourced knowledge in this research context.

To take a step further and extend the above dynamics of knowledge integration to value creation the complementary-asset theory will be the foundation. Developing a new-to-market service or product alone is not sufficient for commercial success without all other parts of the supply chain (manufacturing, marketing, distribution, etc.), and related assets, argues Teece (1986).

Possessing domestic collaboration partners can provide these resources facilitating adaptation. Examples for these domestic partnerships can be suppliers, who can take part in scaling and production, customers can serve with first-hand information about demand and facilitate adoption, or consultants, that can provide specific financial, organizational, or distribution expertise. Recent literature on commercialization similarly shows that network actors help firms overcome adoption barriers and facilitate commercialization capabilities (Salehi et al., 2026), while more process-oriented research highlights the importance of partner engagement and exploitation decisions throughout the innovation journey (Livieratos et al., 2026).

These foundational theories point to three separate hypotheses. The first investigates the net association between foreign breadth and new-to-market innovation, and whether this relationship turns counterproductive. The second aims to reveal if domestic collaboration breadth has the potential to improve interpretation and application of heterogenous foreign inputs. Finally, the third concerns whether the possible anchoring effect of domestic collaboration breadth is stronger regarding commercial performance instead of only the introduction of the market novelty.

2.2 Foreign Collaboration Breadth and New-to-Market Innovation: An Inverted-U Logic

At the initial stages of collaboration development, each new partnership type should raise the possibility of the subsequent introduction of a market novelty as each category presents a potentially unique, new knowledge pool. Foreign customers can provide a fresh point of view on emerging trends from a market that might display different preferences than locally. Foreign suppliers might offer specialised components, or new technologies and know-how in upstream industries. Foreign competitors can contribute with market specific strategic knowledge, while consultants and research institutions transfer knowledge, technologies and network assets. A broader portfolio of partnerships, thus, provides a wider set of knowledge for recombination, by this, raising the likelihood of at least one external source will be beneficial for the

introduction of a market novelty (Leiponen & Helfat, 2010; Hsieh et al., 2018; Kesidou et al., 2023).

A further benefit of foreign breadth is portfolio flexibility. Innovation has the element of uncertainty, therefore, it requires the combination of external knowledge domains instead of a single input. Consequently, wider collaboration portfolio can increase the above-described combinations benefiting firm ability to exploit foreign knowledge domains while also reduces dependence on any one partner. Recent literature is consistent with this logic that a more diverse collaboration portfolio, both functionally and geographically, can enhance innovation processes by enriching knowledge reservoirs and expand access to additional resources (Izotova & Bolívar Ramos, 2024). Jacob et al. (2023) further discusses the benefits of international knowledge sourcing but emphasizes that this is especially true when the portfolio does not exceed over a critical level which the firm can still effectively manage (Jacob et al., 2023).

However, the marginal benefits of extending foreign collaboration breadth should decline over a certain number of partner categories. As further collaboration types are added, knowledge pools can overlap, while costs arise from communication, contracting, monitoring, conflict resolution, and the attempts of objective alignment. Managerial capacity also needs to be divided between more domains, and employee task complexity rises as well with a wider collaboration portfolio. Foreign collaboration partners add more aspects to governance complexity as they can pose institutional, geographical, linguistic, or cultural barriers in a partnership. These do not just raise coordination costs but create cognitive overload possibly leading to the fragmentation of innovation projects. Previous research explicitly highlight limits to the over expansion of foreign collaboration categories, as a geographically diversified portfolio can cause managerial complexity and knowledge redundancy (Audretsch & Belitski, 2023; Jacob et al., 2023). Similar findings are presented by Izotova and Bolívar Ramos (2024)

that both functional and geographic collaboration expansion becomes counterproductive regaining radical innovation.

From these two mechanisms the assumed flipped parabolic, inverted-U relationship of foreign collaboration breadth and new-to-market innovation arises. Until a certain inflection point, the marginal contribution of additional knowledge variety and recombination opportunities should exceed the marginal costs of managing the portfolio, thus, obtaining new foreign collaboration types should improve the probability of the introduction of a market novelty. However, when foreign breadth reaches the optimal level, the marginal coordination costs of a new external partner equal the marginal benefits, and additional collaboration can become counterproductive as coordination and integration demands continue to accumulate. The defined inflection point need not to be universal as companies differ many factors. Therefore, the proposed arguments on knowledge extension and organisational overload conceives the following hypothesis:

H1: Foreign collaboration breadth has an inverted-U-shaped relationship with subsequent new-to-market innovation.

2.3 Domestic Market Collaboration and Local Knowledge Anchoring

According to the inverted-U shape argument, firm capacity to take advantage of foreign knowledge is not automatic. The pivotal element of uncertainty is whether the foreign ideas can be translated into domestic market contexts where the firm seeks to introduce novelty (Rodríguez et al., 2022; Kesidou et al., 2023). Domestic market collaborations can moderate this relationship by helping firms apply and translate foreign knowledge and identify which foreign inputs are worth pursuing as they possess detailed information of local market conditions (Un & Rodríguez, 2018; Kesidou et al., 2023). Thus, increasing domestic market-facing partnerships should make the existing level of foreign breadth more productive (Un & Rodríguez, 2018; Audretsch et al., 2025).

The knowledge selection and translation ability of domestic collaboration might be the most influential mechanism of this relationship, as it makes it possible for specific knowledge or know-how to transfer effectively across markets. Leckel et al. (2022) argues that reducing distance can enhance trust between collaborators, it fosters knowledge exchange and the application of externally sourced knowledge. Although domestic collaboration does not entail geographic proximity, it does provide similar market context with the underlying mechanism that geographically distant knowledge becomes easier to implement when companies function in shared contexts with repeated interaction. Through this mechanism domestic market collaboration can serve as an anchor for global knowledge variety, matching it to local implementation requirements (Bathelt et al., 2004; Un & Rodríguez, 2018).

Another angle where domestic collaboration practices can make the firm more prone to easier knowledge integration is by developing relational routines. As companies learn to collaborate through collaboration, having domestic partners already means a certain level of experience in information sharing, coordination practices, negotiation, and problem resolution. Hsieh et al. (2018) argues that firms having experience with a local collaboration category make them more likely to form foreign collaboration in this category as well, showing the transferability of existing collaboration routines.

However, prior evidence suggests a cautious approach towards this complementarity. According to Haus-Reve et al. (2019) this complementarity takes up a negative value between supply-chain and scientific collaborations. Similarly, Hsieh et al. (2018) show that the complementarity between local and foreign partners varies between partner categories, thus, a uniform positive effect is questionable. According to Kesidou et al. (2023), the geographic scope of collaboration is not uniform for market novel innovation introduction and commercialization. Theoretical grounds for expecting substitution exists as well. Domestic market-facing partners belong to the same categories the foreign measure counts. Thus, once a

firm sources from foreign customers and suppliers, adding their domestic equivalents may repeat knowledge, while using up coordination and absorptive capacity that both portfolios depend on (Cohen & Levinthal, 1990; March, 1991). Although these studies argue that domestic collaborations can become substitutes, this thesis aims to narrow that argument by stating that complementarity should arise as the way domestic market partners contribute is essentially and qualitatively different from the geographically diverse collaborations. Thus, the second hypothesis is the following:

H2: Domestic market collaboration positively moderates the relationship between foreign collaboration breadth and subsequent new-to-market innovation, such that greater domestic market collaboration makes the marginal association more favourable and shifts the point of diminishing returns to a higher level of foreign breadth.

2.4 Domestic Market Anchoring and Commercialization Performance

The anchoring mechanism described in hypothesis 2 should have relevance for both the introduction and commercialization of new-to-market innovation, however it is possible that the assumed effect is different in volume. Commercial success requires additional factors and activities over solely the successful introduction of a market novelty. These consist of refining a value proposition, achieving reliable production adapted to user expectations, marketing activities, distribution and further support of adoption. West and Bogers (2014) arrive to this clear distinction between obtaining, integrating and finally commercialising externally sourced innovation, emphasizing that the success in the first two stages does not necessarily determine substantial value capture in the third. Furthermore, recent commercialization study by Salehi et al. (2026) show that collaboration networks have a substantial role in offsetting adoption barriers, activating downstream workflow and generating market returns.

The complementary-asset perspective further describes the need for domestic market collaborations to capture value. This requires services like manufacturing, sales, distribution, marketing and others along the supply chain, serving as complementary assets that allow the innovator to appropriate value from an invention rather than merely create it (Teece, 1986; Salehi et al., 2026). Companies not possessing these assets can miss out on commensurate returns on their introduced innovative solution, which collaborators could provide distinct from financing and technical invention (Åstebro & Serrano, 2015).

This distinction is the key why moderation is assumed to be stronger for commercial turnover than for sole market novel introduction. The acquisition of foreign knowledge has the potential to contribute to the creation of a market novelty even with limited local market anchoring if the company has enough capacity to recombine and implement this knowledge successfully. However, after introduction the commercial success is more heavily dependent on the level local integration of the novelty to fit demand, the ability to produce and distribute reliably, and on partnerships that can support adoption and repeated sales. Thus, domestic collaboration should help to increase the commercial productivity of foreign knowledge through connecting it to local market-facing assets. Consistent with this logic Kesidou et al. (2023) shows that while extending foreign breadth can help increasing the likelihood of introduction of a market novelty, regarding commercialization, reducing geographic scope can prove to be more beneficial. More recent evidence suggests that successful commercialization does not solely depend on specific partner involvement but also on how collaborative relationships are connected to value capture and exploitation in the long run (Livieratos et al., 2026). Therefore, the third hypothesis is the following:

H3: The positive complementarity between foreign collaboration breadth and domestic market collaboration is stronger for the commercial performance of new-to-market innovation than for the initial introduction of a market novelty.

3. DATA

3.1 Data Description

This thesis uses firm-level panel data from the Scientific Use Files of the Mannheim Innovation Panel (MIP), conducted annually by the ZEW - Leibniz Centre for European Economic Research since 1993. This dataset constitutes the contribution of Germany to the European Commission's Community Innovation Survey. The survey is designed to be representative of the German firm population, it covers firms from manufacturing, mining, energy and water supply, waste management, construction, business-related services and distributive services. The panel construction collects repeated information on the same firms, however, the sample is refreshed periodically compensating for firms exiting the market. The Scientific Use Files are anonymised with selected quantitative variables perturbed, grouped or truncated to protect respondent confidentiality (Peters & Rammer, 2023; ZEW - Leibniz Centre for European Economic Research, n.d.).

The panel design employed in this thesis is non-experimental, solely observational. Instead of random assignment the firms' collaboration patterns and innovation outputs were explored through the survey. Therefore, the results should be treated as associations instead of causal effects as omitted variable bias (OVB), reversed causality or self-selection can arise even when predicting innovation outcomes with previous-year collaboration variables enhances temporal ordering.

The panel has a firm-year structure, however some collaboration indicators are available in only every odd year arising from the consideration to keep the length of the survey limited with rotating question types. Consequently, the unit of analysis is a more specific firm-transition observation, in which the collaboration breadth and control variables of firm i in prediction wave t are connected to the upcoming period's innovation outcome in $t + 1$. Because of data

integrity reasons, the data covers six annual waves from 2019 to 2024, as MIP definitions of product and process innovation were revised post-2019 following the fourth edition of the Oslo Manual (OECD/Eurostat, 2018). The data includes 30 267 firm-year observations from 12 474 distinct firms, with no duplicates. The average observation length per firm is 2.43 years while the median firm is observed for 2 years, which gives an unbalanced panel. From the 12 474 unique firms, only 863 shows up in all six years consistently, while 5076 only appear in a single year.

As the geographically distant collaboration variables are only available in the annual waves of 2019, 2021, and 2023, these are used as predictor waves for the 2020, 2022, and 2024 innovation output variables. This also indicates that a firm does not have to appear in all six observed years but at least one relevant predictor-outcome pair. Therefore, the lags provide a one-wave-ahead survey measure, however, that does not necessarily indicate an innovation to occur exactly one year ahead. In the MIP survey the questions refer to the annual survey wave's reference period, thus not indicating exact dates of starts of collaboration, or innovation introduction.

3.2 Sample and Missing Data

The analysis follows a one-year temporal structure. It begins with the 14 863 firm-year entries from the predictor years, 2019, 2021, and 2023, that is only eligible for analysis if in the following year it has corresponding innovation outcome information. In case of missingness in subsequent observations, the one-wave-ahead outcome will be treated as missing.

The main introduction sample includes observations on foreign and domestic market collaboration breadth, firm size, export intensity and export truncation, R&D intensity and R&D truncation, industry categories and public funding. The last, public funding, is a critical

one as it can cause joint determination or further spurious outcomes, thus, the are estimated across specifications both with and without public funding.

Table 1. Construction of the analytical samples

Sample-selection stage	Firm-wave observations	Unique firms where applicable
Full panel 2019 - 2024	30,267	12,474
Predictor waves: 2019, 2021 and 2023	14,863	—
Observed subsequent introduction outcome	8,511	—
Complete subsequent outcome and export controls	7,768	—
Complete focal variables, R&D, funding and industry controls	4,693	3,519
Subsequent new-to-market innovators within the introduction sample	584	—
Final commercialisation sample with observed turnover category	576	462

Note: The eight firms lost between the innovation-introduction and commercialisation conditions introduced a market novelty but lacked an observed subsequent turnover-share category.

As Table 1 shows that the final introduction, firm-wave sample remains with 4 693 transition observations from 3 519 unique firms. Most firms, 72.66%, only contributes to one transition, 27.33% contributes to at least two transitions, meaning that the average firm contribution is 1.33 observations.

The final commercialization sample is narrower with 576 transitions from 462 firms. This occurs because only those firms constitute this sample that actually introduced a market novelty having the ability for revenue generation and this turnover-share outcome is also observed. As some firms are repeated across samples standard errors are clustered by firm to account for within-firm dependence.

Missing values were kept as missing rather than replaced with zeroes, as innovation questionnaires filters usually allow companies to skip non-applicable questions. For innovation introduction variables however, missing values were changed to zeroes in case the firm was classified as non-innovator, or it reported no product innovation activity. Similarly, the commercialization output variable is turned to zero when there was no market novelty

introduced to begin with. These transformations were employed so zeroes clearly represent established absence of the outcome instead of covering unspecified missingness. In the case of the collaboration variables, the breadth is constructed by summing up different collaboration categories. If all categories are missing, the sum is interpreted missing as well, while if any collaboration category is present, the binary partner indicators are summed. This prevents a completely unknown collaboration portfolio to be interpreted as no collaboration.

Table 2. Missingness in predictor-wave observations

Variable or variable group	Missing	Missing %	Observed	Observed %
Subsequent introduction (ntm_f1)	6,352	42.74	8,511	57.26
Subsequent commercialization (comm_f1)	6,376	42.90	8,487	57.10
Export intensity and truncation indicator	1,609	10.83	13,254	89.17
R&D intensity, logged R&D and truncation indicator	6,462	43.48	8,401	56.52

Note: The denominator is the 14 863 observations in the three predictor waves. Outcome missingness primarily reflects the absence of an immediately following observation or missing outcome information. The R&D transformation and truncation indicator have the same missingness as the underlying R&D variable is the same.

After these recodes, the models use complete - case analysis, without any statistical imputation. This keeps structurally missing outcomes, preventing their replacement with invented values, and ensures the model specifications to use the same observations. However, substantial sample attrition is present as 43.48% of R&D observations is missing from the predictor-wave, which can bias estimates.

3.3 Variables and Measures

Consistently with innovation-measurement research, and following the Oslo Manual, this thesis also distinguishes between the innovation outcome of the introduction of a market novelty and the actual commercial success it generates. It is important to emphasize that these two outcomes are not equivalent as a product can meet the definition of a market novelty without generating economically meaningful sales.

The primary dependent variable is subsequent new-to-market innovation dummy. It is constructed from the original market novelty dummy and completed with survey responses or

routing information, turning it zero when evidence shows no new product introduced. An additional one-wave ahead market novelty variable is constructed as well, in a way that 2019 records contain the company's 2020 innovation outcome, and the same for the 2021 - 2022 and 2023 - 2024 years. The binary market novelty measure is used instead of a product - innovation dummy, as the study aims to investigate product innovation that is new to the concerned market not solely to the firm. Although, this should not be mistaken for radical innovation output as no measure establishes technological discontinuity.

The other relevant dependent variable is the commercialization measure of the introduced market novelty. This variable is constructed from the original variable presenting the market novelty generated proportion of turnover, described by Table 3.

Table 3. Constructed measure of Turnover

Code	Turnover share from market novelties
0	0%
1	More than 0% but less than 5%
2	5% to less than 10%
3	10% to less than 15%
4	15% to less than 20%
5	20% to less than 30%
6	30% to less than 50%
7	50% to less than 75%
8	75% to 100%

Note: Levels of *comm_ord* ranging from 0 to 8. Showing what range of turnover was generated by the market novelty.

Subsequently, a one-wave ahead version of this above-described turnover measure was generated as well. To get the final binary commercialization measure this predictive turnover measure was transformed into a dummy depending on the level of turnover. When the predictive turnover level is above 5 then it is identified as a high turnover novelty and low turnover below. It therefore identifies market novelties accounting for at least 20% of turnover. Although this measure can be sensitive to the chosen threshold and suffer information loss, it

has an intuitive economic meaning that is easy to comprehend and visualize. However, results should be supported with the original ordered outcome. The complementary-assets argument of Teece (1986) further supports the use of a revenue-based outcome measure, as capturing economic value requires different factors than introduction alone.

The main independent variable is foreign collaboration breadth, constructed by summing the ten binary foreign breadth indicators: enterprises within the firm's group, private-sector customers, public-sector customers, suppliers, competitors, consultants or commercial laboratories, universities, public research organisations, industry associations or related organisations, and other foreign partners. The measure created gives equal weight to each category representing the variety of foreign collaboration categories rather than the number of partners. Thus, it ranges from 0 to 10.

This specification logic is consistent with research on external-search as the number of unique knowledge-sources is used to capture the vastness of a firm's search portfolio. Findings of Leiponen and Helfat (2010) show that the likelihood of innovation success can be increased with access to more categories of foreign partnerships, while Laursen and Salter (2006) show an inverted-U relationship, meaning that the benefit of foreign breadth diminishes after a certain number of collaborators. The testing of this inverted-U relationship next to the linear coefficient of foreign breadth requires a squared term as well.

Domestic collaboration breadth is used as a moderator and consists of the count of domestic market-facing partner types: private-sector customers, public-sector customers, suppliers, competitors, consultants or commercial laboratories. Adding up to 5 categories. This variable is narrower than the above-described foreign breadth measure as it aims to serve as a local market anchor instead of a general degree of domestic openness. This logic is consistent with the local-buzz & global-pipelines perspective of Bathelt et al. (2004), meaning that local

collaborations help interpret the non-redundant knowledge that was acquired through foreign partnerships. Therefore, domestic market breadth only includes market-facing categories to provide local validation for hypothesis 2 and 3. Additionally, these two measures should not be handled on the same scale, their general coefficients should not be compared in a linear manner. For the moderation term, it is constructed simply by the product of the foreign breadth and domestic market breadth variables. For more specific information on exact variable names and construction, consult the Appendices.

Controls are added following the logic of previous empirical literature. Firstly, firm size, measured by the number of full-time-equivalent employees. The logarithmic transformation of this variable was necessary to correct for the substantially right-skewedness, while retaining zeroes. Firm size is an important control as it influences the company's resource intensity for innovation, however, larger size can also give rise to bureaucracy and loss of efficiency. Furthermore, according to previous research firms' capacity for innovation and different R&D activities does depend on scale amongst other factors (Cohen & Klepper, 1996).

Export intensity is calculated by dividing foreign turnover with total turnover. Through export mechanisms, companies are more exposed to international competition, customers, and knowledge sources which can consequently boost their innovation activities. Export intensity is a crucial control factor as internationally active companies tend to be more likely not just to collaborate but to innovate as well, thus, this variable can be a potential confounder if not controlled for (Salomon & Shaver, 2005). A further export control variable is export truncation that indicates whether the percentage of exports is affected by the MIP truncation procedure. This procedure anonymises unusually high values (above the 99th percentile threshold) 2019 onwards for anonymity purposes.

Internal R&D intensity is constructed to measure internal R&D expenditure relative to turnover in the following way.

$$\ln_rd = \ln(1 + 100 \times \text{internal R\&D expenditure})$$

The logarithmic transformation is needed to compensate for skewness. Missing R&D information is treated as missing instead of zero as missingness does not indicate the absence of R&D activities. Controlling for R&D intensity helps to differentiate between the firm's exposition to external knowledge and its internal capacity to invest in this knowledge. Similarly to export truncation, R&D truncation also identifies values affected by the MIP truncation procedure, representing a data-adjustment control rather than a theoretical predictor.

Public innovation funding is a crucial control variable that gives the main robustness specifications. It indicates whether the firm received financial support from public institutions for innovation purposes. As monetary support can relax financial constraints it can have a positive influence on innovation processes. At the same time, financial support is not randomly assigned. Promising firms, projects are more likely to be selected and companies struggling are more inclined to apply for these subsidies. Therefore, the robustness check models with and without public innovation funding are useful to verify the main association, as funding may also be jointly determined with collaboration activity. Bronzini and Piselli (2016) provides evidence through a quasi-experiment that that public R&D support has the ability to increase innovation output, however, effects vary by firm size and outcome.

There are two classifications of industry effects. Detailed industry categories and another for a broader four-sector classification. The four-sector classification was applied consistently across specifications to maintain comparability between the introduction and commercialization models. This aims to control for sector-specific differences, such as technological opportunities, product attributes and life cycles, different organizational structures or demand conditions. The categories' sole purpose is to control as their coefficients will represent difference from the omitted sector category instead of being an indicative association.

Furthermore, the models include predicted waves, year indicators for 2019, 2021, and 2023. The purpose of these is to absorb differences across transitions regarding macro conditions, which is uniquely important as this period was disrupted by the COVID-19 pandemic and the subsequent adaptation to the new economic environment.

4. METHODOLOGY

As previously specified, for the empirical analysis pooled firm-level panel data is used from the German Mannheim Innovation Panel (MIP). As certain collaboration measures are solely collected in odd years, three one-wave transitions are examined, where 2019 collaboration measures are linked to 2020 innovation outcomes, 2021 measures to 2022 outcomes, and 2023 measures to 2024 outcomes. Employing these lags in the models establishes temporal ordering, although, as several MIP questions refer to activities in the preceding three years, the survey reference periods may partly cover the same collaboration and innovation episode. Thus, the estimates are treated as associations rather than, causal effects.

The main estimator for the binary introduction of a new-to-market innovation is pooled logistic regression. The commercialization outcome is measured on an ordered turnover-share scale and is thus analysed using ordered logistic regression. All model setups use survey-wave and industry fixed effects, and standard errors are clustered at the firm level to account for repeated observations of the same firm. The control variables are the following: firm size, export intensity, internal R&D intensity, the corresponding truncation indicators, and public innovation funding. As public funding can possibly be endogenous or explain partial variation in the innovation process, each main model is also estimated excluding this variable.

The principal estimates pool all three transitions, while, as a holdout validation the 2023 - 2024 transition is additionally estimated separately. Conditional firm fixed-effects logit and correlated-random-effects specifications could have been used as robustness checks, but as

fixed-effects logit is expected to exclude firms whose innovation status does not change over time, it reduces the sample under a critical level making it unassessable.

4.1 Descriptive Statistics

In the following, descriptive statistics is presented to summarize the applicable variables. Table 4 shows the introduction sample and the commercialization sample. The descriptive values differ as the introduction sample consists of 4 693 firm-wave observations while the commercialization sample only includes 576 observations, only keeping the observations that indicate market novelty.

Table 4: Descriptive statistics of the estimation samples

Variable	Introduction Sample				Commercialization Sample			
	Mean	Median	SD	Min–Max	Mean	Median	SD	Min–Max
New-to-market innovation	0.124	0	0.330	0–1	—	—	—	—
High commercialisation ($\geq 20\%$ turnover)	—	—	—	—	0.196	0	0.397	0–1
Commercialisation category	—	—	—	—	2.755	2	1.904	0–8
Foreign collaboration breadth	0.170	0	0.710	0–10	0.491	0	1.078	0–8
Foreign breadth squared	0.533	0	4.043	0–100	1.401	0	4.927	0–64
Domestic-market collaboration breadth	0.340	0	0.837	0–5	0.762	0	1.139	0–5
Foreign $\times$ domestic-market breadth	0.284	0	1.840	0–36	0.807	0	2.610	0–24
Log firm size	3.523	3.387	1.447	0–7.122	4.066	4.037	1.509	0–7.122
Export intensity	0.160	0.001	0.253	0–1	0.313	0.225	0.298	0–1
Log R&D intensity	0.584	0	0.974	0–4.615	1.449	1.345	1.153	0–4.615
R&D truncation indicator	0.004	0	0.064	0–1	0.009	0	0.093	0–1
Export truncation indicator	0.010	0	0.102	0–1	0.035	0	0.183	0–1
Public innovation funding	0.294	0	0.456	0–1	0.450	0	0.498	0–1

Note: For binary variables, the mean represents the proportion of observations equal to one. The squared and interaction terms are included because they enter the regression specifications directly. Industry and predictor-

wave fixed effects are categorical controls and are summarised separately below rather than by their numerical category codes.

The following Table 5 provides a more informative description on sample composition in percentages, as industry and year fixed effects are used.

Table 5: Sample composition

Sample characteristic	Introduction sample	Commercialization sample
Predictor wave		
2019	27.0%	37.3%
2021	29.8%	30.7%
2023	43.2%	31.9%
Broad sector		
Research-intensive industry	19.8%	42.0%
Other industry	40.8%	33.7%
Knowledge-intensive services	24.1%	19.6%
Other services	15.3%	4.7%

Note: The introduction regressions additionally control for the more detailed industry classification where specified; the four-sector classification is shown here to provide a concise picture of sample composition.

The above tables show that the introduction of a market novelty is relatively uncommon, only 12.4% of observations. Furthermore, the variation between prediction waves also turns out to be substantial as the incidence declines from 17.2% in the 2019-2020 wave to 12.9% in 2021-2022 and even lower to 9.2% for 2023-2024. Because of this variation the inclusion of predictor-wave fixed effects is also supported, assuming different economic conditions under these transitions.

The explanatory variables have a strong concentration towards zero with 91.2% of entries does not record foreign collaboration while 81% does not record domestic collaboration. Observations that constitute of both domestic and foreign collaboration categories are solely 6.1% of total entries, however, among companies that successfully introduced a market novelty the rate of collaboration is higher, 24.1% use at least one foreign partner category and 40.3% use at least one domestic-market partner category. According to Hsieh et al. (2018), this is not an uncommon phenomenon as international relationships tend to be concentrated to a minority

of companies while domestic partnerships are generally more common. Nevertheless, the sparse upper tail of foreign breadth needs to be handled with caution. Although it ranges from 1 to 10, 99% of the observations equals or are lower than 3. Consequently, estimates of high levels of foreign breadth are based on few observations, indicating that predictions close to the upper boundary should therefore be interpreted cautiously.

Truncation indicators are relatively scarce, around 1% for export and 0.4% for R&D truncation in the introduction sample. They are retained as technical controls to account for the censoring of the MIP, thus, to their coefficients there should be no interpretation assigned. Further, two variables, firm size and R&D intensity are right-skewed, which is corrected by their logarithmic transformation.

Regarding representativeness, the MIP was designed to represent the German firm population, extensively covering industries and size classes. However, the sample used is an unweighted, complete-case subsample of this dataset for the sake of the necessary model calculation, thus it cannot be stated that it is representative of all German firms.

Comparing the two narrowed samples, 19.8% of introduction sample while 42% of the commercialization sample is research intensive with also more export-oriented and larger firms found in the commercialization sample. This is an expected pattern as the commercialization sample selection is dependent on the introduction of a market novelty. Furthermore, ZEW's population-level Innovation Survey reported 6.4% of German firms offering market novelties in 2021, while the introduction sample shows 12.4%. Although these two pieces of data are not directly comparable regarding sample construction and different periods, they still reinforce the sample's relative innovation-intensiveness.

In summary, descriptive statistics does not suggest any obvious errors in the data. It identifies two concerns, substantial zero concentration of the collaboration measures, and expected high

standard errors at higher levels of foreign breadth. These do not invalidate the models, however it raises the necessity for careful interpretation, especially at extreme values of foreign breadth.

4.2 Model Specifications

The first hypothesis aims to prove the inverted-U relationship between foreign collaboration and subsequent new-to-market innovation. For this, the following model is used.

$$\text{logit}[P(Y_{i,t+1} = 1)] = \alpha + \beta_1 F_{it} + \beta_2 F_{it}^2 + \gamma X_{it} + \delta_t + \theta_j$$

Here, F represents foreign collaboration breadth, X is the vector of controls, δ_t denotes survey-wave fixed effects, and θ_j denotes industry fixed effects. Support for Hypothesis 1 requires a positive linear coefficient, a negative squared coefficient, and an estimated turning point within the observed range of foreign collaboration breadth. The turning point is calculated as:

$$F^* = -\frac{\beta_1}{2\beta_2}$$

Hypothesis 2 tests whether domestic market collaboration moderates the association between foreign collaboration breadth and subsequent new-to-market innovation:

$$\text{logit}[P(Y_{i,t+1} = 1)] = \alpha + \beta_1 F_{it} + \beta_2 F_{it}^2 + \beta_3 D_{it} + \beta_4 (F_{it} \times D_{it}) + \gamma X_{it} + \delta_t + \theta_j$$

A positive interaction would be consistent with complementarity, whereas a negative interaction would be more consistent with substitution, knowledge overlap, or additional coordination costs. Because interaction coefficients in nonlinear models do not directly represent probability-scale interaction effects, Hypothesis 2 is interpreted using predicted probabilities and average marginal effects at different levels of domestic collaboration.

Hypothesis 3 proposes that the complementarity between foreign collaboration breadth and domestic market collaboration is stronger for commercialization than for initial introduction.

It is assessed by comparing two models.

The commercial significance of new-to-market innovation was originally measured using an ordered turnover-share variable ranging from 0 to 8. An ordered-logit specification was initially considered because this measure preserves the ranking of commercialization outcomes. However, diagnostic testing rejected the proportional-odds assumption for foreign collaboration breadth, its squared term, domestic-market collaboration breadth, and their interaction. A generalized ordered-logit specification was subsequently estimated, but it generated invalid negative predicted probabilities for 129 of the 576 observations, indicating instability associated with the small conditional sample and sparse outcome categories. The nine-category ordered-logit model was therefore not used as the principal commercialization specification.

Instead, commercialization was operationalized as a binary indicator of high commercialization as described before. Market novelties generated at least 20% of the firm's subsequent turnover correspond to commercialization categories 5-8, and it is zero when the turnover share is below 20%, corresponding to categories 0-4. This threshold thus reflects an economically interpretable level of commercial significance rather than being selected according to coefficient significance. Because commercialization can only be observed meaningfully among firms introducing a market novelty, the model was estimated conditionally on high-commercialization and on the availability of the subsequent commercialization measure. This produced a sample of 576 firm-wave observations given by 462 firms.

The conditional high-commercialization model is specified as:

$$\begin{aligned} \text{logit}[P(H_{i,t+1} = 1 | CNTM_{i,t+1} = 1)] = \\ = \alpha + \lambda_1 F_{it} + \lambda_1 F_{it}^2 + \lambda_2 D_{it} + \lambda_3 (F_{it} \times D_{it}) + \gamma X_{it} + \delta_t + \theta_j \end{aligned}$$

where $H_{i,t+1}$ indicates that market novelties generated at least 20% of subsequent turnover, F_{it} denotes foreign collaboration breadth, and D_{it} denotes domestic-market collaboration breadth. The squared foreign-breadth term retains the nonlinear functional form identified in the introduction analysis, while the interaction term tests whether domestic-market collaboration conditions this relationship. X_{it} contains firm size, export intensity, internal R&D intensity, and the associated truncation indicators. The models also include predictor-wave and broad four-sector industry fixed effects, and standard errors are clustered at the firm level.

The specification excluding public innovation funding is treated as the principal model because funding may be jointly determined with collaboration and innovation activity or may lie partly on the pathway between them. A second specification includes public funding as a sensitivity check.

Variance inflation factors were assessed for all specifications to check for multicollinearity issues. All results stayed below the conventional threshold of 10, with a mean VIF of 2.12 and a maximum of 5.76 for foreign collaboration breadth, which follows from its correlation with its own squared term. Detailed industry indicators produced substantially higher values, thus the broad four-sector classification is used throughout all model specifications.

5. RESULTS

The previously examined assumptions, no perfect multicollinearity, the correct functional form of conditional log-odds function, and complete independence across firms, allow for the defined binary-logit specifications. A quadratic term was added to the models to better model nonlinearity, while the functional form of the continuous controls was examined separately. Firm clustered standard errors make sure to account for dependence among repeated firm-level observations. As all models are logit specifications, they do not require normally distributed residuals, there is no assumption requiring homoskedasticity. As clustered standard errors do

not address endogeneity, and unobserved omitted variable bias, or confounding can arise, the modelled relationships are interpreted as associations rather than causal effects.

Hypothesis 1 investigates if the association between foreign collaboration breadth and new-to-market-innovation takes up an inverted-U shape rather than being consistently positive. If this theorized relationship exists, it will entail that initially expanding the diversity of foreign collaboration partners is associated with higher likelihood of introducing a market novelty. However, after a certain inflection point of foreign collaboration breadth, the benefits can diminish. As the collaboration network gets too extensive, it results in increasing search, coordination, governance, and knowledge-integration costs that may eventually outweigh the benefits of accessing diverse foreign knowledge. Pooled binary-logit models were run to examine Hypothesis 1 Table 6 presents these results.

Table 6: H1 Model Specifications with and without Public Funding

Variable	Model 1.0: Excluding public funding	95% CI	Model 1.1.: Including public funding	95% CI
Foreign collaboration breadth	0.525***(0.131)	[0.269, 0.782]	0.514***(0.131)	[0.257, 0.770]
Foreign collaboration breadth squared	−0.081***(0.026)	[−0.131, −0.031]	−0.079***(0.026)	[−0.129, −0.029]
Log firm size	0.219***(0.042)	[0.137, 0.302]	0.219***(0.043)	[0.135, 0.302]
Export intensity	0.500**(0.229)	[0.051, 0.950]	0.514**(0.229)	[0.065, 0.964]
Log R&D intensity	0.688***(0.055)	[0.581, 0.796]	0.672***(0.057)	[0.561, 0.783]
R&D truncation indicator	−1.322*(0.782)	[−2.856, 0.211]	−1.313*(0.781)	[−2.843, 0.217]
Export truncation indicator	0.129(0.379)	[−0.614, 0.873]	0.122(0.378)	[−0.620, 0.862]
Public innovation funding	—	—	0.123(0.114)	[−0.100, 0.345]

Variable	Model 1.0: Excluding public funding	95% CI	Model 1.1.: Including public funding	95% CI
Year 2021	−0.491*** (0.113)	[−0.712, −0.269]	−0.484*** (0.113)	[−0.705, −0.262]
Year 2023	−0.589*** (0.113)	[−0.810, −0.368]	−0.586*** (0.113)	[−0.808, −0.365]
Constant	−2.805*** (0.233)	[−3.262, −2.349]	−2.840*** (0.236)	[−3.304, −2.377]
Observations	4,693		4,693	
Firm clusters	3,519		3,519	
Wald (χ^2)	450.21*** (12 df)		450.43*** (13 df)	

Note: The dependent variable is subsequent new-to-market innovation. Entries are logit coefficients, with firm-clustered robust standard errors in parentheses and 95% confidence intervals in adjacent columns. Broad four-sector industry effects are omitted from the table for readability. The omitted predictor wave is 2019. * $p < .10$, ** $p < .05$, *** $p < .01$.

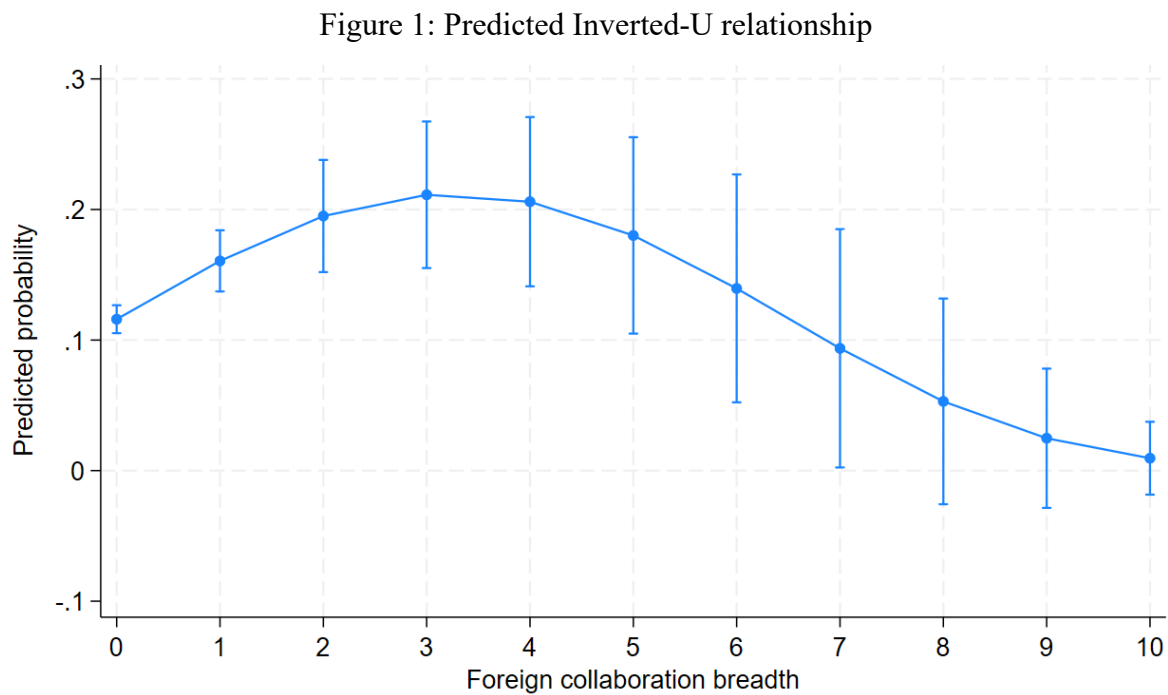
The first specification excludes public innovation funding because of its risk of being jointly determined with the firms' innovation and collaboration activities, or even its possible role as a catalyst on the way between collaboration and subsequent new-to-market innovation output. Thus, this specification is treated as the principal model. The second specification includes public funding to assess the sensitivity of the findings conditional on its omission. As Table 6 shows, the results were highly stable across the two specifications, the foreign-collaboration coefficient solely changed from 0.525 to 0.514 the quadratic coefficient from −0.081 to −0.079, and the estimated turning point from 3.26 to 3.25 partner categories, see calculation below.

$$F^* = -\frac{\beta_1}{2\beta_2} = -\frac{0.525}{2(-0.081)} = 3.26 \rightarrow -\frac{0.514}{2(-0.079)} = 3.25$$

Public funding was itself statistically insignificant with a p-value of 0.279. Therefore, the evidence supporting an inverted-U association was robust to the alternative treatment of public funding. This pattern describes the before discussed association of foreign collaboration breadth with first increasing, then decreasing likelihood of introducing a new-to-market

innovation. Nevertheless, because of the substantial absence of observations at high levels of foreign collaboration the descending part of the relationship should be interpreted with caution.

For the principal specification Wald tests confirms the significance of both the linear and quadratic foreign-collaboration breadth terms. The linear term differed significantly from zero, $\chi^2(1) = 16.13$, $p < .001$, while the squared term was also significant, $\chi^2(1) = 10.00$, $p = .002$. The positive linear coefficient and negative quadratic coefficient supports the suspected inverted-U-shaped association between foreign collaboration breadth and subsequent new-to-market innovation.



Note: Predicted probabilities with 95% confidence intervals from the principal model.

After examining the predicted probabilities, Graph 1 clearly shows the inverted-U shaped association, further strengthening the results. First, the predicted probability of introducing a market novel product rises from 11.6% at zero foreign partner categories to 16.1% at one, 19.5% at two, and it peaks approximately at 21.1% with three different categories of foreign collaboration partner. The further extension of foreign partner categories makes the probability decline to 18.0% at five categories, 14.0% at six, and 9.4% at seven, drawing out the inverted-

U shape association. However, as confidence intervals are widening substantially at higher foreign collaboration breadth levels because of event sparsity, the decline can only be interpreted as an association not causality.

As a temporal robustness check, all models are re-estimated in the last transition period to test temporal stability of the of results and their independence of COVID-related or post-COVID transitions. Hypothesis 1 was run using only the 2023-2024 transition using only the 2,028 observations of that period. The positive linear term, $\beta = 0.670$ with a p-value of 0.002, and negative squared term with $\beta = -0.132$, and a p-value of 0.016, remained statistically significant, with an estimated turning point of 2.55 partner categories. Predicted probabilities further showed the inverted-U pattern, thus providing temporal support of the robustness of Hypothesis 1. Export truncation was omitted automatically because of multicollinearity issues, this however, does not invalidate the robustness check.

In summary, the findings suggest that the access to diverse foreign knowledge through a network of foreign collaborators initially benefits innovation, but extensive collaboration breadth may eventually add additional coordination costs that outweigh the initially created benefits. Thus, the findings represent an association instead of a causal effect. This interpretation follows recommended practices for evaluating inverted-U relationships combining coefficient signs, the turning point, and predicted probabilities (Haans, Pieters & He, 2016).

Hypothesis 2 aims to test whether domestic market collaboration positively moderates the relationship between foreign collaboration breadth and new-to-market innovation. Specifically, this positive moderation would entail that innovation benefits resulting from broader foreign collaboration is facilitated by the increase in domestic-market collaboration breadth. Assumingly, domestic partners would take on a role to complement and to interpret foreign

knowledge and help to translate it into innovations suited for the domestic market. To estimate this relationship, a pooled binary logistic regression was used, the quadratic term was kept, and a moderation interaction term was added, broad four-sector industry fixed effects, and predictor-wave fixed effects. See results in Table 7. below.

Table 7: H2 logistic regression results: moderation by domestic-market collaboration

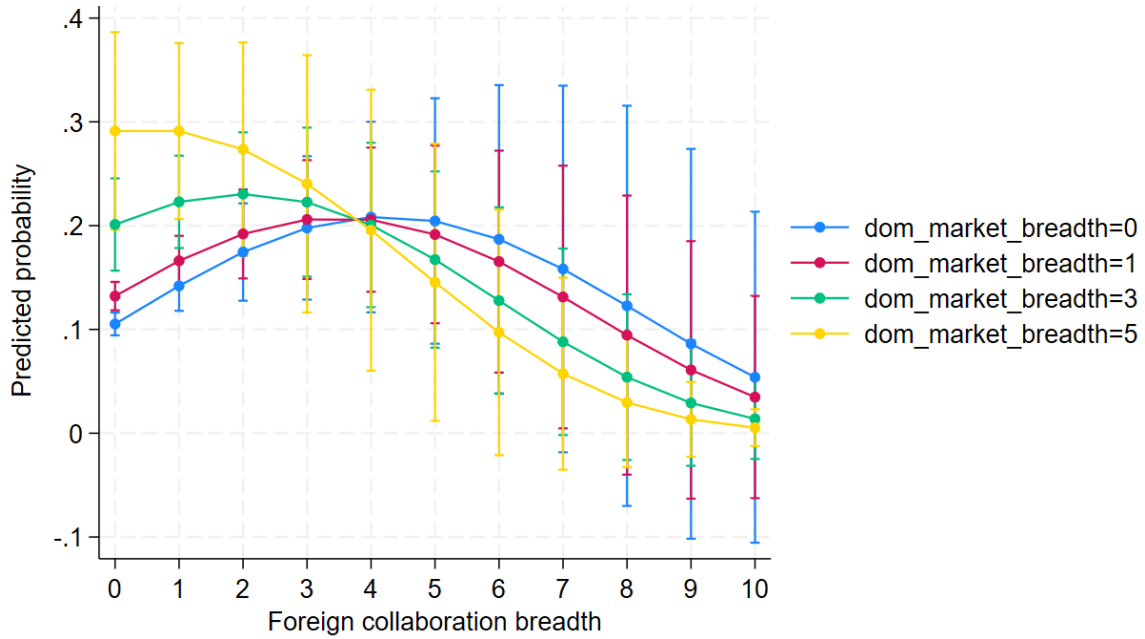
Variable	Interaction Model 1.0.: Excluding public funding	95% CI	Interaction Model 1.1.: Including public funding	95% CI
Foreign collaboration breadth	0.447***(0.136)	[0.181, 0.713]	0.445***(0.136)	[0.179, 0.711]
Foreign collaboration breadth squared	−0.053**(0.025)	[−0.102, −0.003]	−0.052**(0.025)	[−0.102, −0.003]
Domestic-market collaboration breadth	0.297***(0.061)	[0.177, 0.416]	0.294***(0.061)	[0.174, 0.414]
Foreign × domestic-market breadth	−0.079**(0.039)	[−0.155, −0.003]	−0.079**(0.039)	[−0.155, −0.002]
Log firm size	0.205***(0.042)	[0.122, 0.288]	0.205***(0.042)	[0.122, 0.288]
Export intensity	0.596***(0.229)	[0.146, 1.045]	0.599***(0.230)	[0.148, 1.050]
Log R&D intensity	0.637***(0.055)	[0.528, 0.745]	0.632***(0.057)	[0.520, 0.745]
R&D truncation indicator	−1.305*(0.690)	[−2.657, 0.047]	−1.302*(0.691)	[−2.656, 0.052]
Export truncation indicator	0.120(0.374)	[−0.613, 0.854]	0.117(0.374)	[−0.616, 0.850]
Public innovation funding	—	—	0.038(0.115)	[−0.187, 0.263]
Year 2021	−0.479***(0.113)	[−0.701, −0.257]	−0.477***(0.113)	[−0.699, −0.255]
Year 2023	−0.615***(0.115)	[−0.840, −0.390]	−0.614***(0.115)	[−0.839, −0.389]
Constant	−2.825***(0.234)	[−3.284, −2.366]	−2.835***(0.236)	[−3.299, −2.372]
Observations	4,693		4,693	

Variable	Interaction Model 1.0.: Excluding public funding		Interaction Model 1.1.: Including public funding	
	95% CI		95% CI	
Firm clusters	3,519		3,519	
Wald (χ^2)	463.62*** (14 df)		464.01*** (15 df)	

Notes: Entries are logit coefficients, with firm-clustered robust standard errors in parentheses and 95% confidence intervals in adjacent columns. Broad four-sector industry effects omitted for readability. * $p < .10$, ** $p < .05$, *** $p < .01$.

As Table 7 clearly shows the two specifications, with the before established inclusion and exclusion of public innovation funding. It is clear that the results stay highly stable across the two specifications as coefficient volumes and directions do not change substantially. From this, in the following, the principal interaction model will be further analysed, which excludes public funding, to keep the model integrity on a higher level. Further, in the principal interaction model foreign collaboration breadth was positive with a statistically significant coefficient of 0.447, ($p < .001$) and the squared term was negative, also statistically significant with a coefficient of -0.053 , ($p = .037$), retaining the inverted-U pattern when domestic-market breadth equals zero. Moreover, domestic-market breadth also obtained a positive coefficient of 0.297 ($p < .001$) of high statistical significance. However, counterintuitively to Hypothesis 2 the foreign-domestic interaction term came out negative with a coefficient of -0.079 and reached significance at the 5% confidence interval ($p = .043$). As before noted the Interaction Model 1.1. produced nearly identical results, and the model fit also only changed negligibly. Because of the interaction term's negative coefficient, the assumed moderation effect is the opposite to what was hypothesized, consequently, Hypothesis 2 was not supported. However, as logistic model coefficients do not represent probability scale interactions, additional predicted probabilities and marginal effects were examined that can be found in the Appendix in detail. The following Figure 2 shows the dynamic change of predicted probabilities.

Figure 2: Domestic Market Moderation



Note: Lines show predicted probabilities; error bars represent 95% confidence intervals.

Nor predicted probabilities nor marginal-effects analysis provides evidence that domestic-market collaboration strengthened the association between foreign collaboration breadth and new-to-market innovation output. Although, the predicted probability of innovation output rises at zero foreign breadth from 10.5% to 29.1% when domestic breadth extends, the likelihood of innovation falls substantially after foreign collaboration breadth reaches four. On level six of foreign breadth the probabilities are already decreasing as more domestic market types are added, specifically from 18.7% of domestic breadth zero until 9.7% at domestic breadth level five. As marginal effects follow a similar pattern, the evaluated shape is rather a decreasing one than a positive complementarity as proposed in Hypothesis 2. As observations at the higher joint breadth levels were sparse and confidence intervals overlap substantially findings should not be interpreted as conclusive proof but an association. According to this, Hypothesis 2 was not supported, and the pooled model showed a statistically significant negative interaction.

Temporal robustness check for Hypothesis 2 was estimated using the same 2023-2024 sample of 2 028 observations. Foreign breadth continues to be positive, $\beta = 0.549$ with a p-value of 0.015. Domestic-market breadth also remained positive, $\beta = 0.199$ with a p-value of 0.029, while the squared foreign-breadth term was negative but only marginally significant ($\beta = -0.119$, $p = .056$). Crucially, the interaction term was close to zero and non-significant ($\beta = -0.012$, $p = .885$) and remained so with the inclusion of public funding ($\beta = -0.011$, $p = .892$). Consequently, the corresponding margins and predicted probabilities did not indicate systematic moderation either, leaving Hypothesis 2 unsupported, with a temporally unsupported negative interaction.

Hypothesis 3 had the purpose to examine whether the complementarity between foreign and domestic market breadth is stronger for subsequent high commercialization or initial introduction of a market novelty. While the positive complementarity was not supported according to the results of Hypothesis 2, instead they indicated a rather decreasing pattern at the introduction stage, the comparison of Hypothesis 3 can serve with meaningful insights. As the stages of commercial exploitation and innovation development is foundationally different, Hypothesis 3 aims to assess where the foreign–domestic interaction becomes more favourable. Knowing the results of Hypothesis 2, the complementarity of foreign and domestic market breadth is treated as an interaction or an empirical possibility rather than an established relationship.

In preliminary testing of the models the proportional-odds assumption was tested, which rejected the focal predictors as instability was detected in the nine-category commercialization specification. For this, high commercialization, or commercialization success, was defined as market novelties that generate at least 20% of turnover. Following, two pooled binary-logit model estimates, first, the probability of new-to-market innovation, second, the probability that

the innovation is high commercialization, conditional on being a market novel product. Both specifications include broad-sector and wave fixed effects, with firm clustered standard errors.

Table 8: H3 pooled binary-logit models: innovation introduction and high commercialization

Variable	Introduction model	95% CI	High-commercialization model	95% CI
Foreign collaboration breadth	0.457*(0.137)	[0.189, 0.725]	−0.213(0.280)	[−0.763, 0.336]
Foreign collaboration breadth squared	−0.052**(0.026)	[−0.103, −0.002]	0.014(0.049)	[−0.082, 0.111]
Domestic-market collaboration breadth	0.299*(0.060)	[0.181, 0.417]	0.095(0.142)	[−0.184, 0.373]
Foreign × domestic-market breadth	−0.088**(0.039)	[−0.165, −0.011]	−0.009(0.100)	[−0.205, 0.187]
Log firm size	0.202***(0.042)	[0.119, 0.285]	−0.271***(0.095)	[−0.457, −0.085]
Export intensity	0.647***(0.226)	[0.204, 1.091]	0.069(0.463)	[−0.839, 0.977]
Log R&D intensity	0.604***(0.054)	[0.499, 0.709]	0.151(0.123)	[−0.090, 0.393]
Export truncation indicator	0.096(0.371)	[−0.631, 0.823]	0.273(0.649)	[−1.000, 1.546]
Broad-sector fixed effects				
Other industry	−0.376**(0.145)	[−0.660, −0.092]	−0.664**(0.329)	[−1.310, −0.019]
Knowledge-intensive services	−0.759***(0.173)	[−1.099, −0.420]	0.561*(0.322)	[−0.070, 1.192]

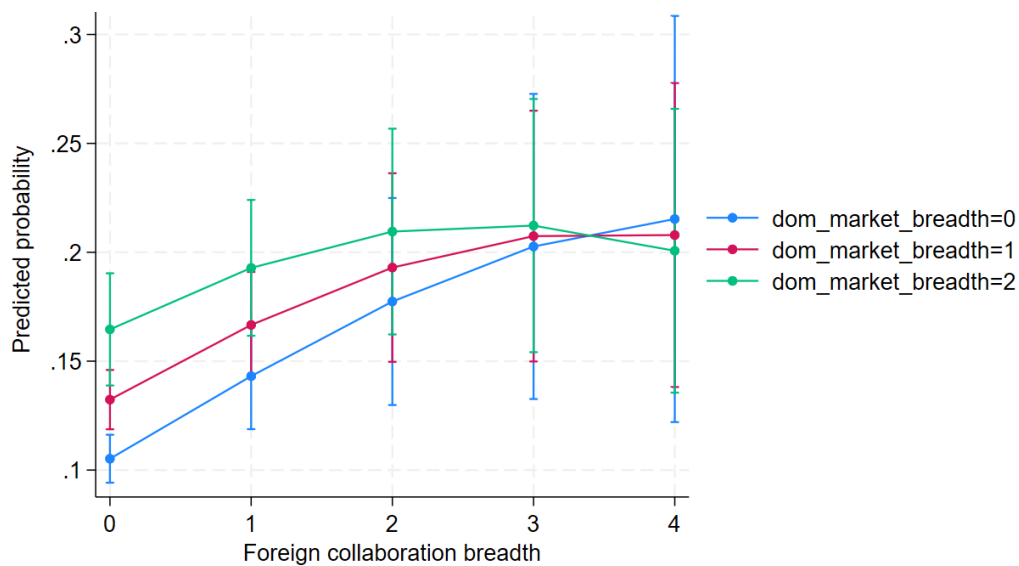
Variable	Introduction model	95% CI	High-commercialization model	95% CI
Other services	−1.085*** (0.231)	[−1.538, −0.632]	0.545 (0.512)	[−0.459, 1.549]
Predictor-wave fixed effects				
Year 2021	−0.475*** (0.113)	[−0.696, −0.253]	−0.327 (0.272)	[−0.861, 0.206]
Year 2023	−0.615*** (0.114)	[−0.840, −0.391]	0.495* (0.263)	[−0.021, 1.011]
Constant	−2.788*** (0.233)	[−3.245, −2.332]	−0.711 (0.482)	[−1.656, 0.234]
Observations	4,693		576	
Firm clusters	3,519		462	
Wald (χ^2 (df))	461.45 (13), ($p < .001$)		54.93 (13), ($p < .001$)	

Notes: Logit coefficients with firm-clustered robust standard errors in parentheses. Both models exclude public innovation funding, omit the R&D truncation indicator because it perfectly predicted high commercialization, and include broad four-sector and predictor-wave fixed effects. Reference categories are omitted. * $p < .10$, ** $p < .05$, *** $p < .01$.

Table 8 continues to show the inverted-U shape between foreign collaboration breadth and the introduction of market-novel innovation, the linear term being positive and significant with a coefficient of 0.457, while the squared term is negative, significant with a coefficient of −0.052. Furthermore, the attenuation pattern also shows up instead of complementarity between domestic market breadth and market novel innovation because of the negative, significant interaction term of −0.088. Contrasting, the High-commercialization model does not show any of these focal terms to be significant including the foreign-domestic interaction term ($b = -0.009$, $p = .927$). In the second, high commercialization sample, all five

truncated R&D-intensity indicators reached high commercialization (above 20% turnover), which resulted in perfect prediction. This made the model drop truncated R&D-intensity as a variable. To maintain the same control structure of the introduction and high-commercialization models truncated R&D-intensity was eventually dropped from both models. It must be further specified that high commercialization events are relatively sparse, 113 (19.62%) exactly, from only 576 conditional commercialization observations, as firms that did not introduce a market novelty cannot meaningfully have turnover from that, and these firms are excluded from the sample by design. Although, high commercialization was a minority outcome, the observations were frequent enough to support binary logistic estimation. Consequently, because the two logit specifications use different outcomes and samples, their coefficients are not suitable for formal cross-model comparison. Thus, Hypothesis 3 is interpreted primarily through predicted probabilities and marginal effects. Therefore, from this initial model domestic collaboration does not seem to strengthen the foreign-breadth relationship in terms of commercialization, thus Hypothesis 3 is not supported.

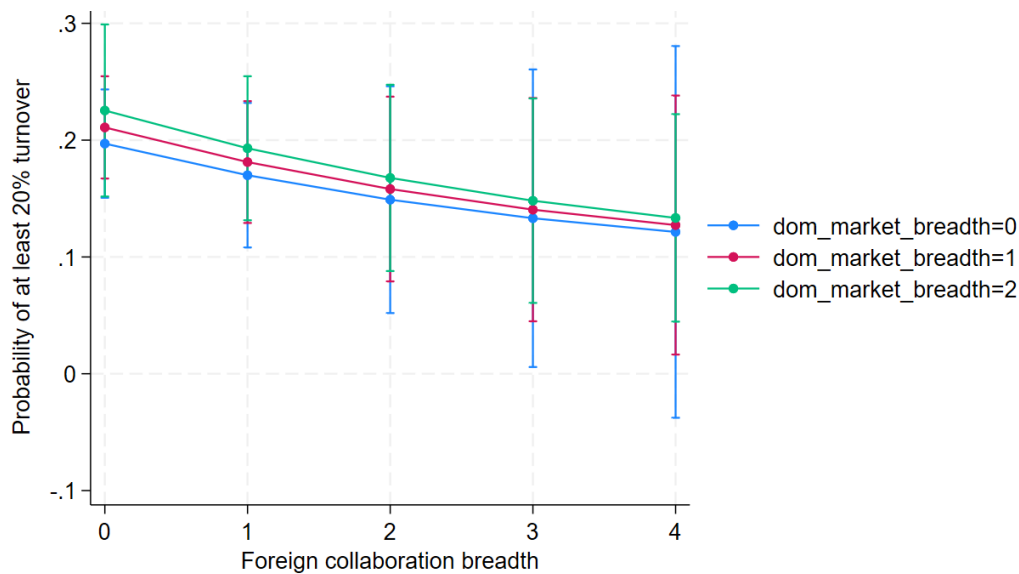
Figure 3.1: Predicted Probabilities of Market-novel Innovation



Note: Predicted probabilities of the pooled binary logit introduction model. Lines show estimated probabilities of introducing a market novelty across levels of foreign collaboration breadth, separately for domestic-market collaboration breadth = 0, 1, and 2. Error bars indicate 95% confidence intervals.

Figure 3.1 shows that by increasing domestic collaboration breadth, predicted probability levels of introducing a market novelty decreases with the extension of foreign collaboration breadth. The predicted-probability curves converge to approximately 3 to 4 foreign partner categories, which makes it consistent with the negative foreign-domestic interaction term in Hypothesis 2, which describes a diminishing effect instead of a complementarity between domestic and foreign collaboration breadth in terms of new-to-market innovation. As before specified, the overlapping confidence intervals at higher breadth levels require cautious interpretation.

Figure 3.2.: Predicted Probabilities of Conditional High Commercialization



Note: Predicted probabilities from the pooled binary logit commercialization model. Lines show the estimated probability that market novelties account for at least 20% of turnover across levels of foreign collaboration breadth, separately for domestic-market collaboration breadth = 0, 1, and 2. Error bars indicate 95% confidence intervals.

Figure 3.2 depicts 3 seemingly almost parallel curves, and an unambiguous diminishing pattern of predicted probabilities of high commercialization as foreign collaboration extends. Although, as the curves lay close, it does not come with a great effect which level of domestic market breadth is considered. Predicted probabilities decline as foreign breadth increases from approximately 19.7 - 22.5% at foreign breadth 0 to 12.1 - 13.3% by foreign breadth 4. These

results are consistent with the high commercialization model's close-to-zero interaction-term coefficient. Thus, wider domestic collaboration breadth did not complement the foreign-breadth association with market novel innovation meaningfully, therefore, providing no support for Hypothesis 3.

According to the temporal robustness check foreign breadth remained positively associated with the likeliness of new-to-market innovation, $\beta = 0.555$ with a p-value of 0.016, while the squared term was negative but only marginally significant, $\beta = -0.114$ with a p-value of 0.074. The coefficient of domestic-market breadth was positive and significant, $\beta = 0.203$ with a p-value of 0.026, while the interaction term was not substantial and statistically insignificant, $\beta = -0.029$ with a p-value of 0.729. In the conditional high-commercialization sample none of the coefficients were statistically significant, similarly, predicted probabilities also showed little separation across domestic-collaboration levels in the commercialization model. Thus, the temporal robustness check neither supports Hypothesis 3, there is no stronger foreign-domestic complementarity for commercialization than for introduction of market novelty according to the findings.

6. DISCUSSION

The thesis explores the relationship between foreign collaboration breadth and subsequent new-to-market innovation, if domestic market collaboration has a moderating effect on this relationship, and whether this complementarity is more prevalent for commercialization.

The findings show strong support for Hypothesis 1. Foreign collaboration has a flipped parabolic association with new-to-market innovation as the likeliness of the introduction of a market novelty rises initially, and peaks around 2 to 3 foreign partner categories before it starts to decline. This pattern shows up in the temporal robustness check of the last transition wave of 2023 to 2024 further supporting the notion. Thus, the argument, supported by previous

literature, that foreign breadth facilitates the access to non-redundant knowledge diversity, but these benefits are eventually outweighed by rising coordination costs (Laursen & Salter, 2006; Audretsch & Belitski, 2023; Jacob et al., 2023).

Hypothesis 2 was not supported. In contrast with the expected complementarity results, the interaction term between foreign and domestic breadth turned out to be negative and statistically significant in both pooled specifications. Although, domestic breadth alone was positively associated with innovation output, but the increase in domestic market-facing collaboration categories had an attenuating rather than a strengthening association. This aligns with previous literature describing the ambiguous relationship of combining collaboration types, and that the complementarity between them is not universal (Hsieh et al., 2018; Haus-Reve et al., 2019). However, the interaction term being significant and negative may indicate a substitute relationship association instead of the hypothesised complementarity. Although, as the temporal robustness specification did not show significance, the negative moderation should not be interpreted as a temporally stable substitution effect.

Why these two might work against each other instead of complementing one another is worth exploring, since the theory behind Hypothesis 2 expects the opposite. Firstly, one possibility is redundancy. The domestic measure counts market-facing categories, customers, suppliers, competitors and the foreign measure count the same kinds of partners abroad. A company already working with foreign customers and suppliers may find that domestic partners of the same type can add knowledge repetition, the added categories bringing additional coordination complexity without the addition of new knowledge (March, 1991; Jacob et al., 2023). A second possibility is absorptive capacity to be a binding constraint. The same managers and routines have to handle every relationship, and once realized absorptive capacity is stretched, foreign and domestic ties draw on it at each other's expense instead of mutual reinforcement (Cohen & Levinthal, 1990; Zahra & George, 2002). The third concerns the direction in which partners

pull. Domestic market partners lie close to known demands that the firm might already understands, and responding to them tends to favour adaptation over novelty, which is the hypothesized local solution that distant knowledge is supposed to break (Bathelt et al., 2004; Laursen & Salter, 2006). As none of these mechanisms is tested here, and interaction disappears in the 2023-2024 temporal robustness sample, they are better treated as associations worth examining than as definitive conclusions.

Hypothesis 3 was again, not supported. Domestic collaboration breadth failed to strengthen the influence of foreign breadth on commercialization more than for market novelty introduction. This supports the described distinction between creating a new-to-market innovation and capturing value in terms of commercial returns. Collaboration patterns or breadth may not explain the sales performance following the introduction of a market novelty (Teece, 1986; West & Bogers, 2014; Kesidou et al., 2023).

This thesis further contributes to open-innovation literature (Chesbrough, 2003) by establishing that foreign collaboration has limits, defining the optimal approximate number of foreign partner categories, at around three. Although, increasing diversity is only beneficial as long as the firm has capacity to coordinate and integrate knowledge successfully. Further findings implicate that, although domestic collaboration may be valuable independently, but the combination of foreign and domestic collaboration portfolios does not automatically reinforce one another. Empirically, highlighting the distinction between knowledge creation and value capture by treating introduction and commercialization as distinct outcomes is also a key contribution.

Additionally, theoretical contribution goes beyond the location of the turning point where foreign collaboration breadth becomes counterproductive. Open innovation has largely been framed as a question of how openly a company extends partnerships (Chesbrough, 2003), while

these results point to composition as much as degree, since the same number of partner categories performs differently depending on where those partners lie. The findings also place absorptive capacity in a slightly different role. If breadth in one location has the ability to change the worth of breadth in another, absorptive capacity behaves less like a stock that every new tie takes from, but more like a shared constraint (Cohen & Levinthal, 1990; Zahra & George, 2002). Lastly, the local buzz and global pipelines argument treats local and distant knowledge as complements. However, results here suggest that this cannot simply be assumed at the individual firm-level, at least not when the local side is measured through market-facing collaboration (Bathelt et al., 2004).

6.1 Managerial Implications for Collaboration Portfolio Design

For managerial practices the findings implicate that more collaboration is not necessarily better. Companies should prioritize to only extend their collaboration portfolios until the complexity is realistically maintainable, until around two to three foreign partner categories, rooting from the inverted-U association uncovered. Exceeding this number provides a diminishing likelihood of introducing a market novelty. Thus, companies should focus on collaboration breadth portfolios to be managed rather than maximized and evaluate the existing coordination capacity for each additional partner type as a pivotal constraint.

Findings also caution against treating domestic market collaboration breadth as an anchor for foreign breadth. Although, on its own, domestic market breadth is positively associated with the introduction of a new-to-market innovation, their interaction suggests that the two portfolios might be in one competition for managerial capacity rather than reinforcing the integration of one another. As collaboration breadth shows little association with commercialization, firms might benefit more from the strengthening of complementary assets, like production capacity, market access, distribution, customer adoption, or sales activities, instead of widening their collaboration portfolio (Teece, 1986).

6.2 Limitations and Future Research Directions

The results are not without limitations. The findings describe associations rather than causal effects. Even with temporal robustness checks, structurally overlapping survey period references of the MIP questionnaire can limit safe interpretation. Although, using previous-wave collaboration improves temporal ordering, reverse causality, omitted variable bias and self-selection into collaboration cannot be ruled out.

Furthermore, there are additional measurement limitations to consider. Collaboration breadth variables weight every category equally, not including information on quality, duration or depth, thus, companies with identical scores can collaborate completely different manners. As collaboration in general is concentrated on lower breadth levels the upper range of estimates of the inverted-U association is less precise. Regarding the binary measure of commercialization, it also discards information remaining sensitive to the turnover threshold. Overall, internal validity is sufficient, most drawbacks resulting from the dataset, and survey structure.

Regarding external validity, the analysis relies on unweighted, complete-case subsample of firms from Germany, thus the findings cannot be generalized globally but can be handled as a good representation of the German firm population and firms operating in similar national or institutional contexts. Future research could further investigate in this direction. The use of administrative or linked data could identify collaboration effects more credibly, separating partner types instead of aggregation can also give a more specific picture, next to employing measures of depth, quality or coordination capabilities. Commercialization could be tracked by continuous revenue data, and overall, longer horizons of company data could be beneficial where value creation and capture is more likely to materialize. An additional interesting research direction could be to focus on case studies, to evaluate companies' collaboration qualities further.

7. CONCLUSION

In summary, foreign collaboration breadth can support the introduction of market novelty, but its benefits are bounded. Initially, the likelihood of introducing a market novelty increases by extending foreign collaboration categories but becomes counterproductive once the value of additional knowledge sourced is outweighed by cooperation complexity. In addition, the influence of domestic collaboration should be handled independently rather than using it as a local anchor according to the ambiguity of the findings. Although domestic collaboration breadth is positively associated with innovation output on its own, it did not reinforce the adaptation of foreign knowledge, nor offered additional leverage for commercialization. Overall, seeing the results together bring one to the conclusion that instead of focusing on collaboration breadth, the capabilities surrounding it might be more relevant, such as coordination capacity, implementation capabilities, or external knowledge selection strategies.

Thus, successful open innovation does not seem to depend solely on the breadth of collaboration but rather the additional capabilities to effectively select external knowledge, have the ability to exploit that while keeping coordination under control

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APPENDICES

Table 9. Operationalisation of outcome and focal variables

Construct	Source variable(s)	Constructed variable	Coding and role
New-to-market innovation	mneu, filter, pd	ntm	1 = market novelty; 0 only where response or questionnaire routing establishes no market novelty
Subsequent introduction	ntm	ntm_fl	One-wave-ahead binary outcome
Market-novelty turnover	mneup, ntm, filter, pd	comm_ord	Ordered turnover-share category, 0–8
Subsequent commercialisation	comm_ord	comm_fl	One-wave-ahead ordered outcome
High commercialisation	comm_fl	high_comm_fl	1 if category ≥ 5 , corresponding to at least 20% of turnover
Foreign collaboration breadth	koa1–koa10	foreign_breadth	Count of foreign partner categories, 0–10
Quadratic foreign breadth	foreign_breadth	F2	foreign_breadth ²
Domestic market breadth	kod2–kod6	dom_market_breadth	Count of domestic market-facing categories, 0–5
Any domestic market collaboration	dom_market_breadth	dom_market_any	1 if breadth > 0; robustness measure
Foreign–domestic complementarity	Foreign and domestic breadth	FD or model interaction	Product term used to test moderation
Combined innovation outcome	ntm_fl, comm_fl	ntm3_fl	0 = no novelty; 1 = novelty with under 20% turnover; 2 = novelty with at least 20%

Note: Detailed variable construction for dependent variables and main independent variables